

**Absolute Value and Exponential Functions Performance Task****DIRECTIONS**

5 items • 8 points • 45 minutes • Form A

Read the scenario, define quantities, show mathematical work, and explain conclusions. The task is scored with an eight-point rubric.

Equipment Inspection Model Comparison

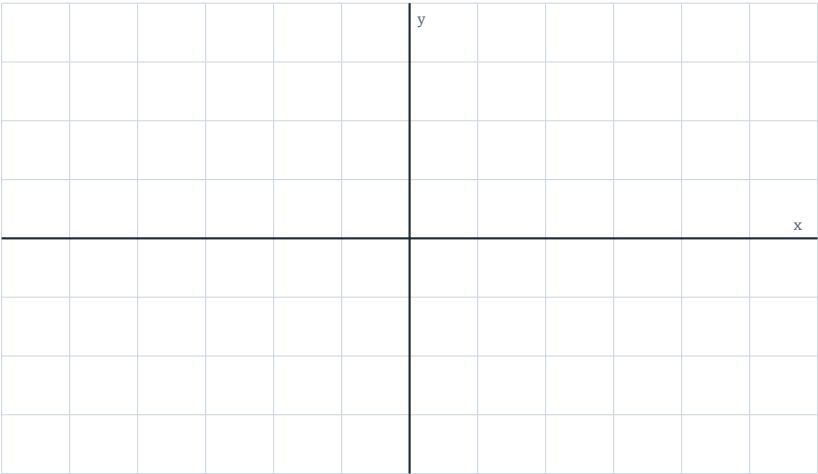
Form A. The equipment inspection model comparison uses $D(t) = |t - 6| + 2$ for a temperature-deviation score and $V(t) = 1200(0.8)^t$ for a dollar-value model, where t is time in years.

1. Identify the vertex, minimum, symmetry, domain, and range of D .

2. Classify V as growth or decay and state its initial value and percent rate.

3. Evaluate both functions at $t = 3$, with correct units.

4. Compare the graphs and rates of change over the modeled domain. Explain why one has a constant multiplicative rate and the other does not.



5. Defend which model is appropriate for deviation and which is appropriate for value. Address why their output units should not be directly compared.

Eight-Point Rubric

Category	Points	Full-credit evidence
Modeling and setup	2	Defines quantities and creates an appropriate mathematical model.
Accuracy	2	Uses correct procedures and obtains accurate results.
Representation	2	Uses and labels an appropriate graph, table, equation, or structural representation.
Interpretation and justification	2	Explains the conclusion, units, constraints, and reasonableness.